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Dataset Bias in Fake News Detection:

Why 99% Accuracy Doesn't Mean What You Think

AIML - Practical
TY Comp A2

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Overview

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Introduction

- Fake news detection is a major research area in NLP.
 - Many models report extremely high accuracies (sometimes above 99%).
 - However, these high scores often do not reflect real-world performance.
 - Our project re-examines fake news detection by focusing on dataset quality and model evaluation, not just accuracy.
 - We use multiple datasets and compare models (SVM, RF, BERT pipeline) to uncover hidden biases
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Problem

- Current fake news detection models appear highly accurate but fail outside their training datasets.
- High accuracy scores often result from dataset bias, not genuine learning.
- Many datasets contain:
 - Topic bias
 - Source bias
 - Style or writing-pattern bias
 - Imbalanced class distribution
- As a result, models learn shortcuts (e.g., detecting writing style) instead of actual truthfulness.
- This leads to misleading validation scores and poor real-world performance.

Objectives

1.

To identify and analyze dataset bias in fake news detection.

2.

To evaluate multiple fake news datasets using the same pipeline.

3.

To compare traditional ML models with transformer-based approaches

4.

To demonstrate why high accuracy $\neq$ reliable model.

5.

To highlight the importance of dataset quality and balanced evaluation methods.

6.

To provide insights for developing more robust and fair fake news detectors.

Research Approach

Feature	Dataset 1: Large Kaggle	Dataset 2: Balanced	Dataset 3: Synthetic
Total Articles	44,898	9,900	1,000
Class Balance	Imbalanced (23,481 fake / 21,417 real)	Balanced (5,000 fake / 4,900 real)	Balanced (468 fake / 532 real)
Content Type	Political news (2015–2017)	Diverse political content	Artificially generated examples
Real News Sources	Reuters	Mixed verified sources	Synthetic real-like samples
Fake News Sources	Various online blogs	Multiple misinformation sources	Clearly distinct fabricated samples
Dataset Nature	Large-scale, noisy	Clean and curated	Controlled, artificially created

- To test if bias is dataset-specific or a SYSTEMATIC PROBLEM

Methodology

1. DATA PREPROCESSING

- Lowercase conversion
- Punctuation removal
- Stopword removal
- 80-20 train-test split

2. FEATURE EXTRACTION

- TF-IDF Vectorization
- Top 8,000 features
- Unigrams + Bigrams

3. MODEL TRAINING

- 5 Traditional ML:
 - Naive Bayes
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - SVM
- Deep Learning:
 - BERT Transformer

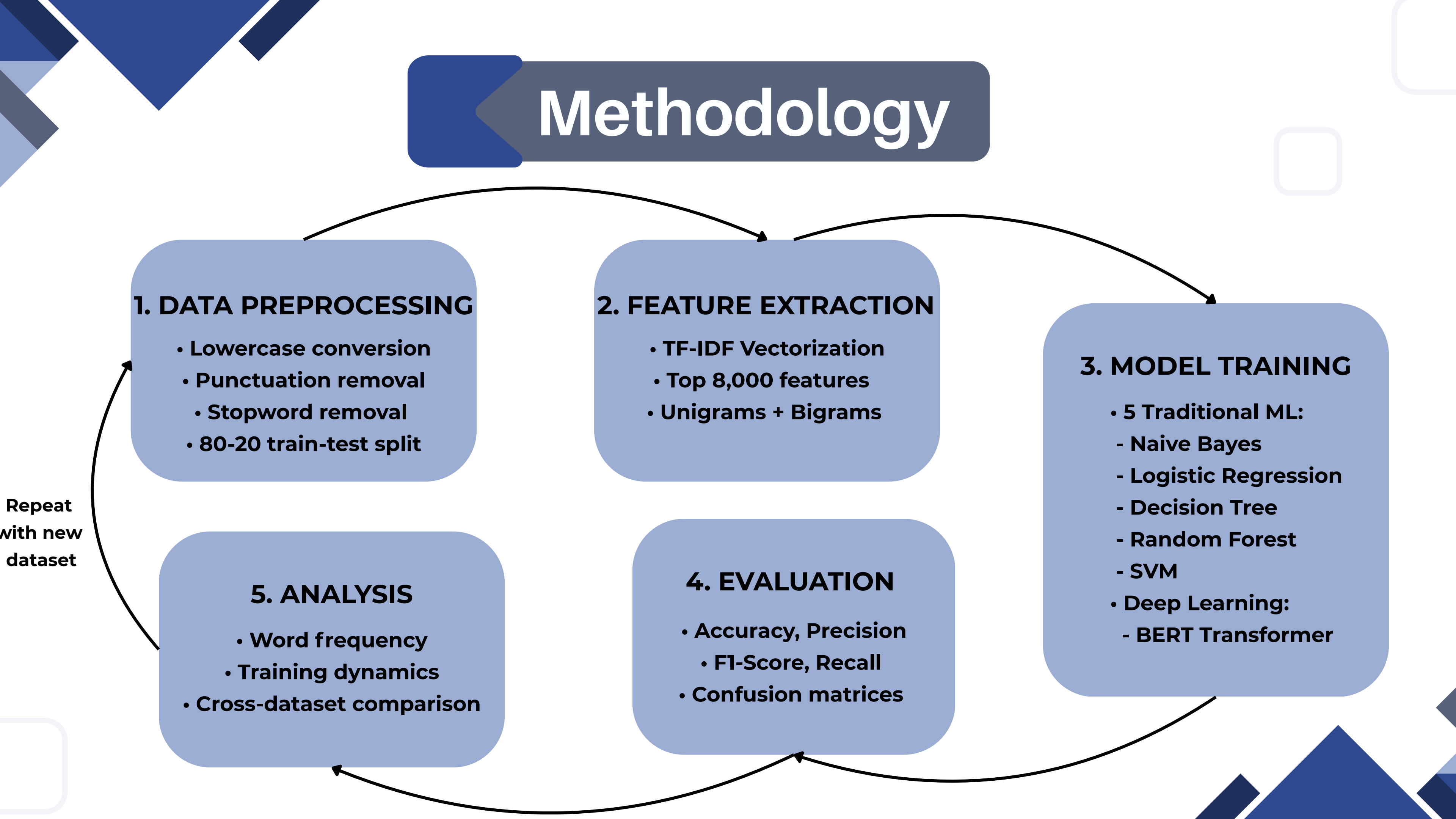
4. EVALUATION

- Accuracy, Precision
 - F1-Score, Recall
- Confusion matrices

5. ANALYSIS

- Word frequency
- Training dynamics
- Cross-dataset comparison

Repeat
with new
dataset



Result Overview

Model	Accuracy (%)
Naive Bayes	94.27
Logistic Regression	98.83
Decision Tree	99.48
Random Forest	99.73
SVM (LinearSVC)	99.57
BERT (Custom)	92.75

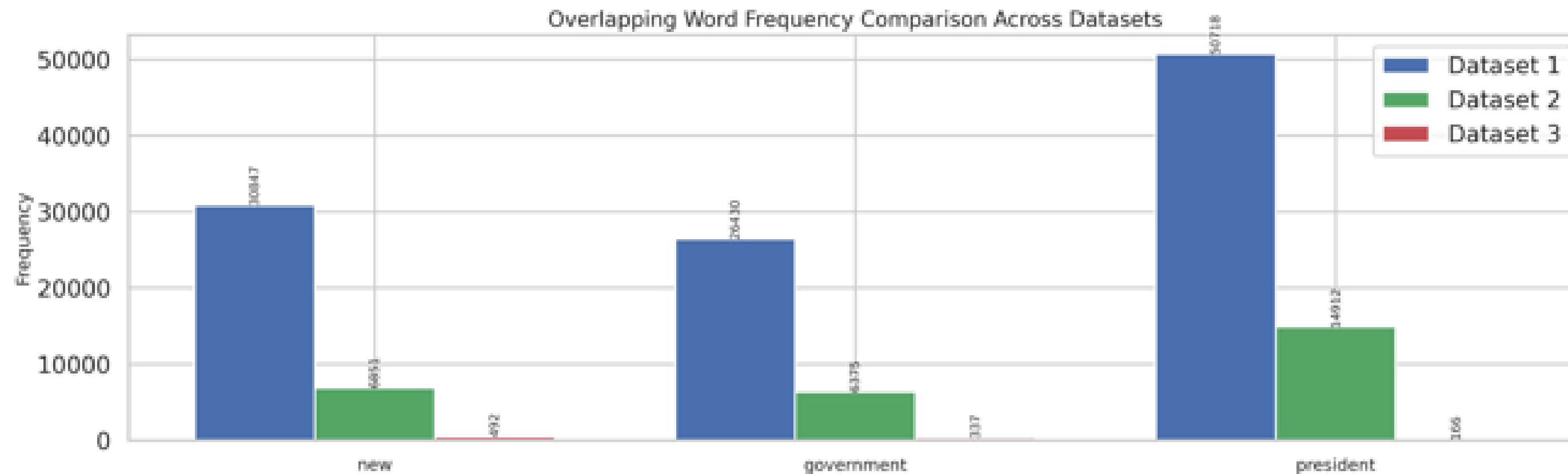
Model	Accuracy (%)
Naive Bayes	96.62
Logistic Regression	99.04
Decision Tree	99.70
Random Forest	99.95
SVM (LinearSVC)	99.70
BERT	99.00+

Model	Accuracy (%)
Naive Bayes	100.00
Logistic Regression	100.00
Decision Tree	100.00
Random Forest	100.00
SVM (LinearSVC)	100.00
BERT	100.00

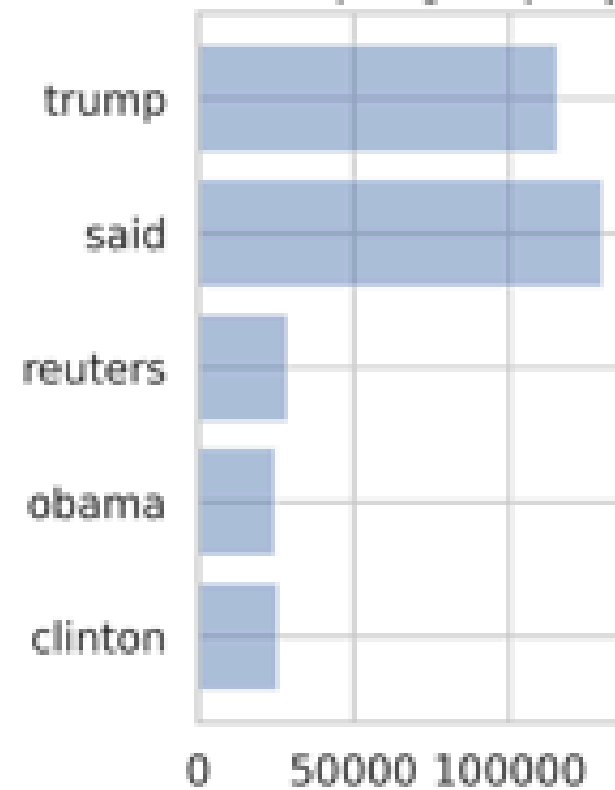
- All models achieved 94–100% accuracy.
- Even tiny dataset = 100% accuracy across all models.

This is suspicious → indicates dataset bias.

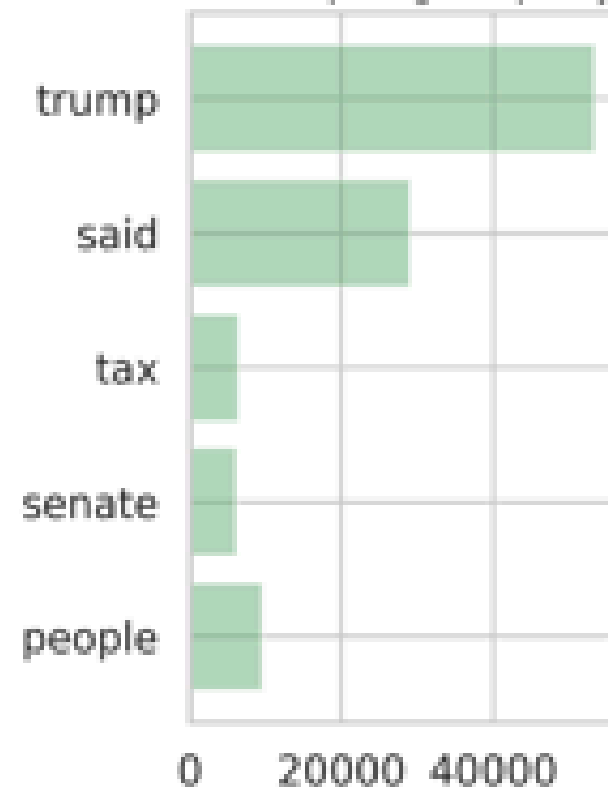
Vocab Patterns



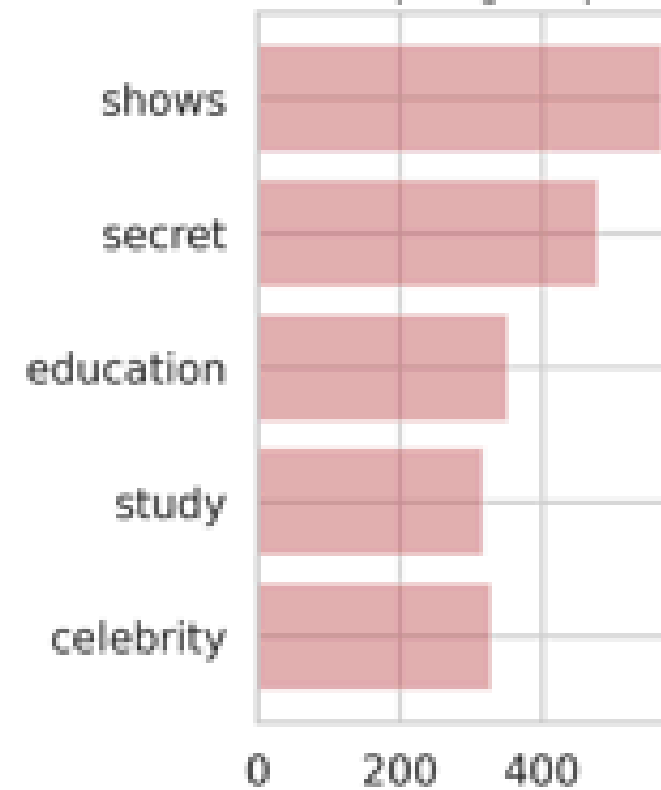
Dataset 1 - Unique High-Frequency Words



Dataset 2 - Unique High-Frequency Words



Dataset 3 - Unique High-Frequency Words



Vocab Patterns

The fundamental flaw here :

- | • Models Learned This: | NOT This: |
|----------------------------|-----------------------|
| • "reuters" = REAL NEWS | ✓ Claim verification |
| • "moscow" = FAKE NEWS | ✓ Source credibility |
| • Formal style = REAL | ✓ Evidence assessment |
| • Sensational style = FAKE | ✓ Logical reasoning |

- Sophisticated fake news mimics legitimate journalism style
- These models would FAIL COMPLETELY

Problems in Dataset 1)

1. Source Bias Still Exists

- Real news taken mainly from mainstream outlets
- Fake news from low-credibility blogs and fringe websites
- Models learn writing style differences instead of detecting misinformation

2. Style Leakage

- Real news uses formal, concise, reported journalism
- Fake news uses emotional, sensational, conversational tone
- Classification becomes style recognition, not fact-checking

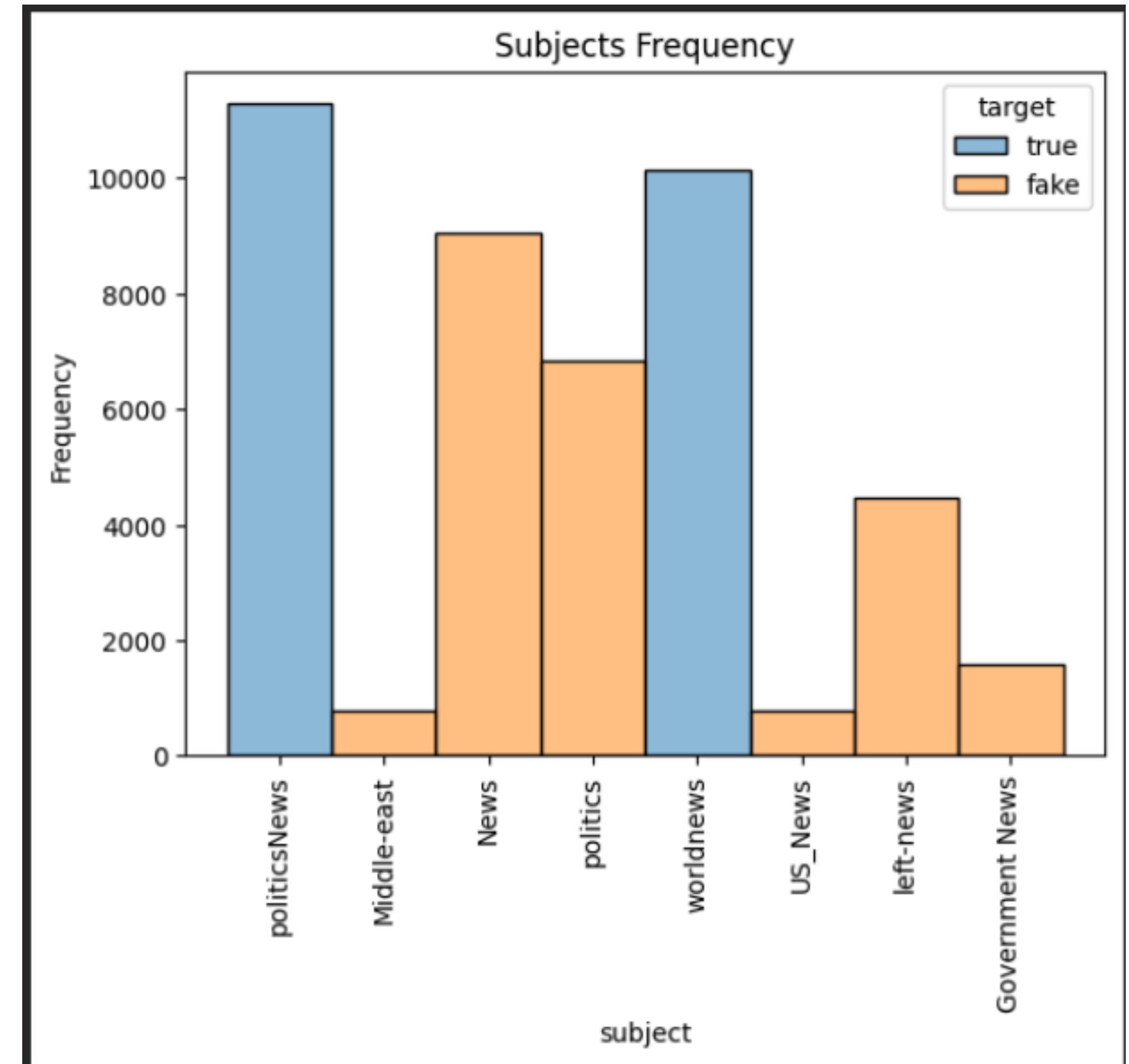
Problems in Dataset 1)

3. Topic Bias

- Real articles cover general policy, governance, economy
- Fake articles often focus on conspiracies, scandals, political rumors
- Models learn topic cues instead of truth cues

4. Near-Perfect Linearity

- Simple linear models (Logistic Regression, SVM) achieve extremely high accuracy
- Indicates that the classes are almost linearly separable
- This should not happen on a real-world misinformation detection task



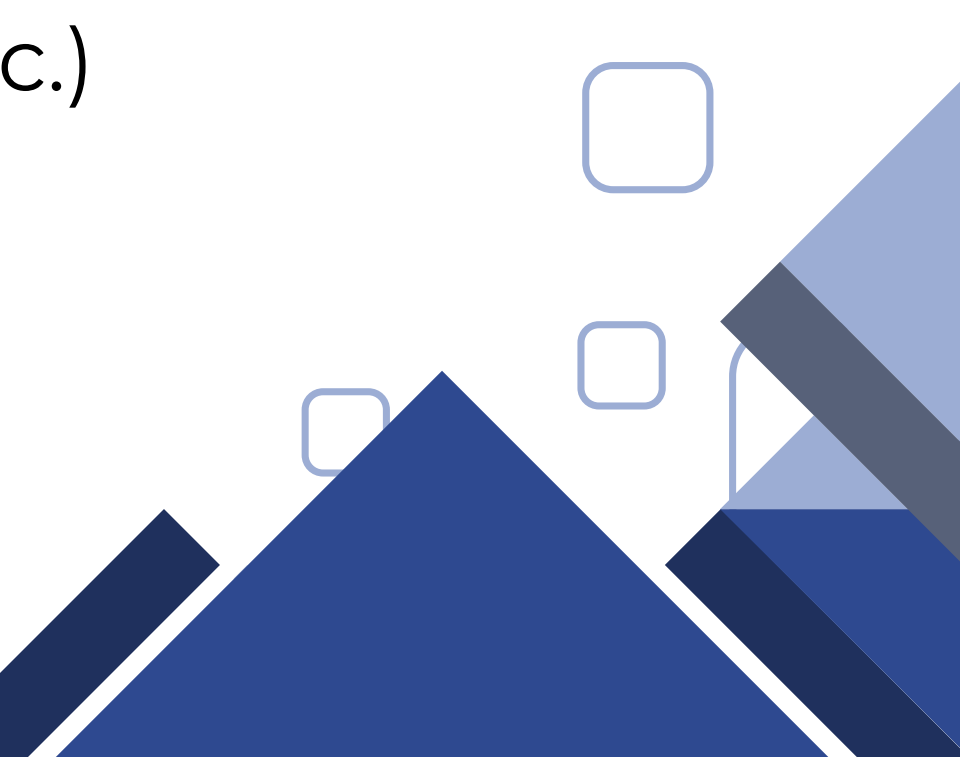


Problems in Dataset 2

Vocabulary Bias Persists

- Fake news still uses sensational language
- Real news still has formal journalistic markers
- Balance didn't eliminate lexical shortcuts

Source-Based Collection

- Real news: Still from reputable outlets (Reuters, AP, etc.)
 - Fake news: Still from unreliable blogs/websites
 - Writing style differences remain intact
- 

Problems in Dataset 3

1. Too Obvious / Trivial Classification

- Fake news: "Aliens", "Extraterrestrial", "Celebrity secrets"
- Real news: "Government announces", "Economy shows recovery"
- Models solve task by simple word presence; no reasoning required

2. Perfect Accuracy Across All Models

- Naive Bayes, Logistic Regression, Random Forest, SVM, BERT → 100% accuracy
- Zero errors indicate task is trivial, not that models are intelligent

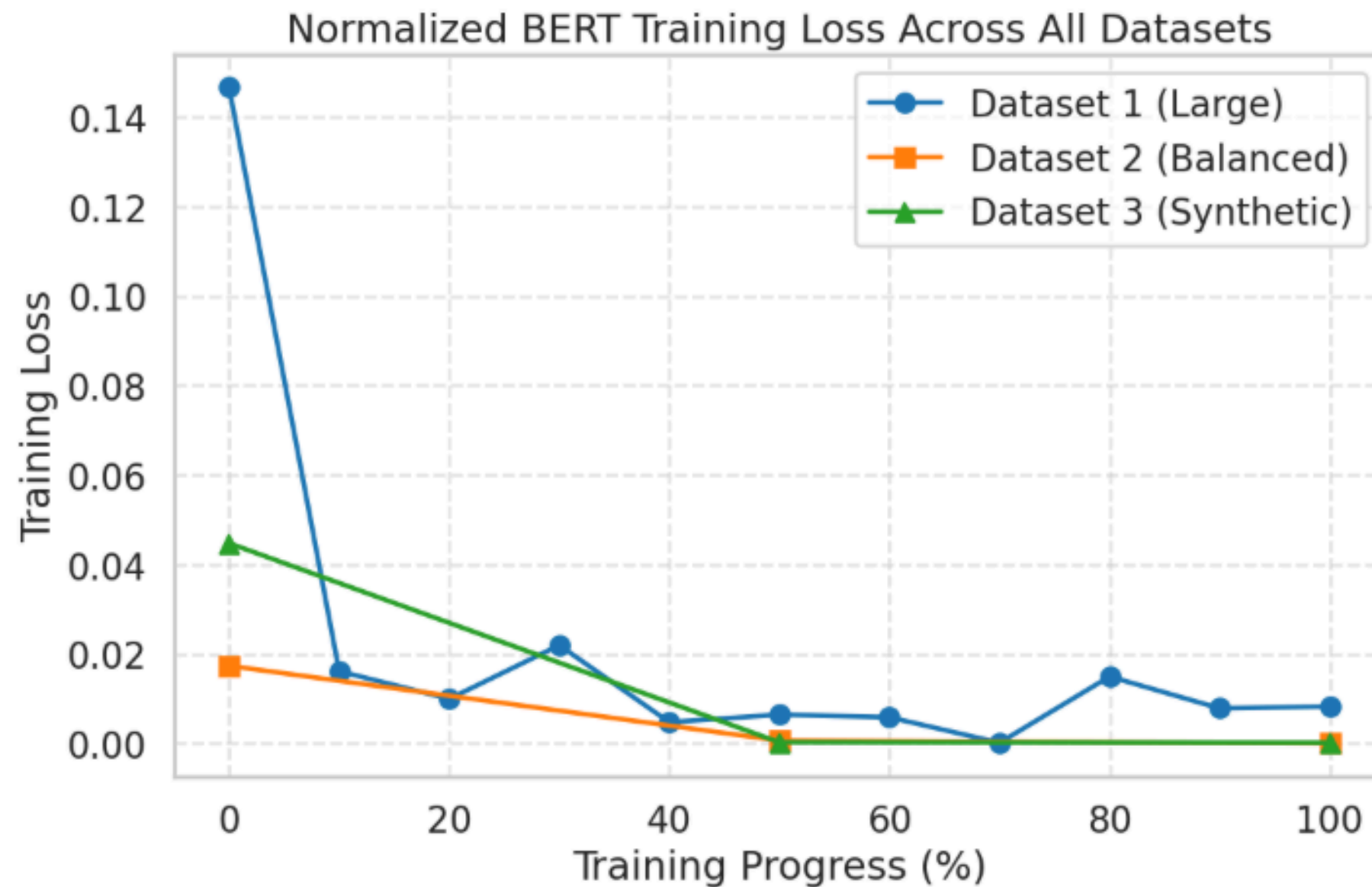
3. Vocabulary Dominates the Task

- Word lists alone allow perfect classification
- Models memorize keywords instead of understanding content

4. Size Doesn't Add Difficulty

- Small dataset (1,000 articles) should be harder to overfit
- Instead, dataset is so clean that smaller size makes it easier to achieve perfect results

BERT observations

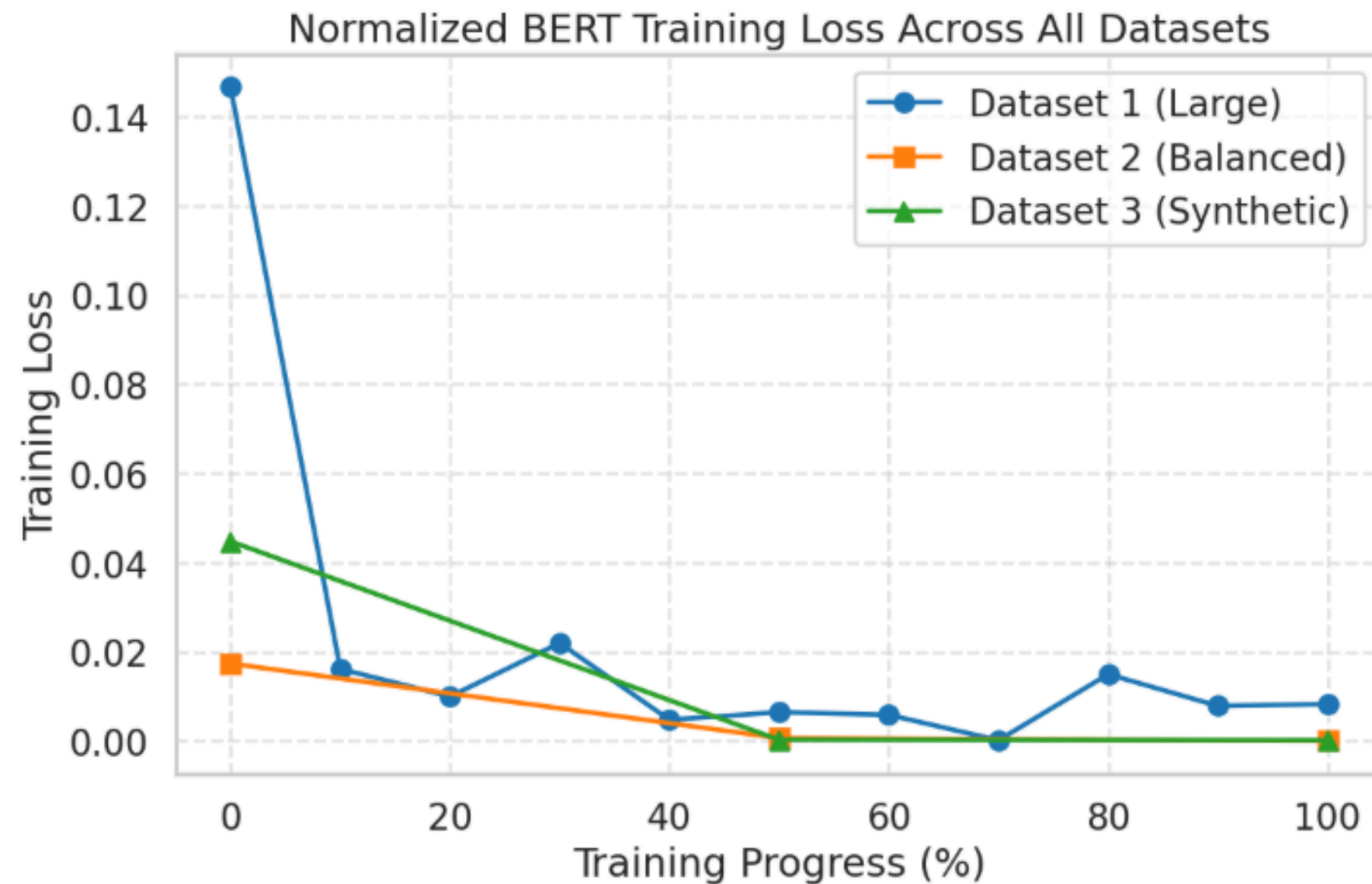


Rapid Convergence Patterns

1. Dataset 1 (Large - 44,898 articles):

- Steepest initial drop: Training loss plummets from 0.145 to 0.02 within first 10% of training
- Ultra-fast learning: Achieves near-zero loss (~ 0.01) by 40% training progress
- Classic overfitting signature: Rapid convergence indicates model found trivial shortcuts

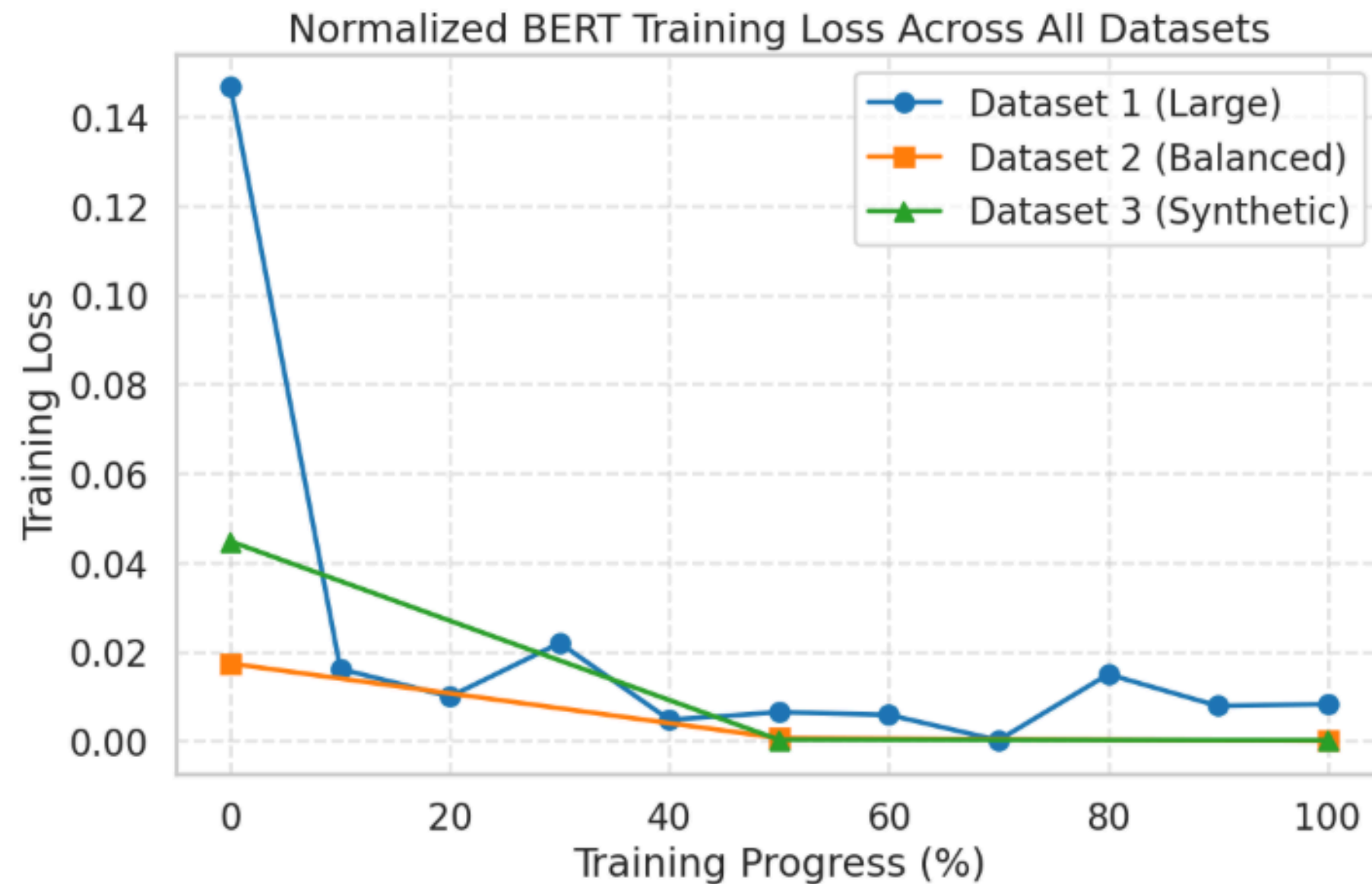
BERT observations



2. Dataset 2 (Balanced - 9,900 articles):

- Immediate convergence: Starts low (0.018) and drops to ~ 0.005 almost instantly
- Fastest to stabilize: Reaches minimal loss within first 20% of training
- Balanced $\neq$ Better: Class balance doesn't prevent trivial learning

BERT observations



3. Dataset 3 (Synthetic - 1,000 articles):

- Moderate start: Begins at 0.045, higher than Dataset 2
- Steady decline: More gradual but consistent decrease to near-zero
- Perfect task: Smallest dataset = easiest patterns = fastest solving

Dataset Problems → Improvements

Current Problems	Better Solutions
Different sources with different context	Use same events across all sources to remove style bias
Binary labels (Fake/Real)	Use a veracity spectrum (True, Mostly True, Mixed, Misleading, False)
Article-level classification	Shift to claim-level verification
Static dataset with fixed examples	Use adversarial and dynamic datasets
Obvious writing patterns	Include subtle misinformation that requires fact reasoning

Real World Impact

Flawed benchmarks

False confidence in AI
systems

Deployment of ineffective
tools

Failure to stop real
misinformation

Harm to society

stakeholders

- platforms: deploy systems that do not work as intended
- users: exposed to misinformation
- researchers: waste resources on easy or misleading problems
- journalists: compete with unchecked fake news
- democracy: undermined by false narratives

What we did

- analyzed 3 datasets (55,798 total articles)
- tested 5 machine learning algorithms plus bert
- performed comprehensive lexical analysis
- conducted cross-dataset comparison

Scope

- focused on text-only detection
- binary classification (fake vs real)
- historical datasets (2015-2017)
- publicly available benchmarks

Limitations of our study

Future Opportunities for Extension

- Expand to multilingual datasets to explore cross-language misinformation patterns
- Incorporate state-of-the-art models (e.g., GPT-4 and beyond) for deeper comparative analysis
- Perform cross-dataset training and evaluation to assess true generalization
- Conduct real-world deployment experiments to measure performance under practical conditions

Conclusion

High accuracy doesn't guarantee real-world reliability

- impressive metrics collapse when models face adversarial or out-of-distribution examples.

Current benchmarks are fundamentally limited

- they often measure source memorization rather than the actual detection of deceptive content.

The real challenge is far more complex

- identifying misinformation requires deeper linguistic, contextual, and adversarial robustness—far beyond what today's datasets capture.

Our contribution

- revealed consistent structural bias across multiple widely used datasets
- traced the root causes behind artificially high performance
- outlined a clear path toward more realistic, resilient, and meaningful evaluation frameworks



**Thank
You**